University of British Columbia, Vancouver
Department of Computer Science

CPSC 304 Project Cover Page

Milestone #: 4

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Group Number: 13

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By typing our names and student numbers in the above table, we certify that the work in the attached assignment was performed solely by those whose names and student IDs are included above.

In addition, we indicate that we are fully aware of the rules and consequences of plagiarism, as set forth by the Department of Computer Science and the University of British Columbia

Final project description

This application functions as a stock market tracker and portfolio management system designed to incorporate relationships between various market components. The system uses operations such as asset acquisitions, personal information updates, and removals while providing many views of information on specific tables that may be useful to the average trader, such as joining multiple tables to view trader holdings and utilizing queries to identify specific sectors or companies that exceed global performance averages, as well as seeing which traders are doing well, and what stock they have. Additionally, tables can be organized and downloaded into a CSV.

Changes since Milestone 2

UNIQUE phone number added to entity People in order to satisfy the UNIQUE constraint of query 2. Additionally removed Buys, Sells, Operates, and Reports, as they were not used and clashed with key constraints for other tables.

State of database after running DDL.sql

Located at the bottom of the page.

SQL Queries

1. Insert

In file: appService.js
Line #: 145

```
INSERT INTO Contains (Email, CreatedDate, TickerSymbol, Quantity,
                      PurchasePrice)
VALUES (:email, TO_DATE(:createdDate, 'YYYY-MM-DD'), :tickerSymbol,
       :quantity, :purchasePrice)
```

2. Update

In file: appService.js
Line #: 169

```
UPDATE People
SET Networth = :newNetWorth, Cash = :newCash, Phone = :newPhone
WHERE Email = :email
```

3. Delete

In file: appService.js
Line #: 205

```
DELETE FROM People
WHERE Email = :email
```

4. Selection

In file: src/router/router.js
Line #: 120

```
SELECT *
FROM Assets
WHERE [Column] = :val0 [AND/OR] [Column] = :val1
```

5. Projection

In file: src/controller/feature-projection.js
Line #: 64

```
SELECT [column_names]
FROM [table_name]
```

6. Join

In file: appService.js
Line #: 239

```
SELECT p.Email, p.Name, p.Networth, c.TickerSymbol, c.Quantity,
       a.CurrentPrice,
       a.Name AS AssetName, (c.Quantity * a.CurrentPrice) AS
       MarketValue, (c.Quantity * (a.CurrentPrice - a.PurchasePrice))
       AS ProfitAndLoss
FROM People p
JOIN Contains c ON c.Email = p.Email
JOIN Assets a ON a.TickerSymbol = c.TickerSymbol
WHERE p.Email = :email
```

7. Aggregation with GROUP BY

In file: src/controller/feature-seven.js
Line #: 5

```
SELECT Exchange, count(*) ,avg(CurrentPrice), avg(CurrentPrice -
       LastPrice)
FROM Assets
GROUP BY Exchange
```

This query groups assets by their respective stock exchanges to calculate the total number of listings, the average current price, and the average price movement for each market.

8. Aggregation with HAVING

In file: appService.js
Line #: 301

```
SELECT Sector, AVG(EarningsPerShare) as AvgEPS
FROM Stocks
GROUP BY Sector
HAVING AVG(EarningsPerShare) > :threshold
```

This query finds sectors that have an EPS above some user defined threshold.

9. Nested Aggregation with GROUP BY

In file: src/controller/feature-nine.js
Line #: 19

```
SELECT Sector, avg(EarningsPerShare)
FROM Stocks
GROUP BY Sector
HAVING avg(EarningsPerShare) >
     (SELECT avg(EarningsPerShare)
     FROM Stocks)
```

This query finds high performing industry sectors by comparing each sector's average earnings per share against the global average for all stocks in the database.

10. Division

In file: src/controller/feature-ten.js
Line #: 5

```
SELECT p.Email
FROM People p
WHERE NOT EXISTS (SELECT c.CompanyID
                  FROM Companies c
                  WHERE c.IsPublic = 1
                  AND NOT EXIST    (SELECT o.Email
                                    FROM Owns o
                                    WHERE o.Email = p.Email
                                    AND o.CompanyID = c.CompanyID))
```

This query finds individuals who maintain an ownership stake in every company designated as public in the database.

Informal Citations

Assuming the milestone guide did not require a formal citation:

https://www.w3schools.com/ - General refresher for HTML, CSS, and JS used for the project. We already had rough previous knowledge on these from courses taken before.

Sample node_project provided at the start of milestone 4 was referenced for certain CSS styles and javascript functions, and often built off of for other functions.

The same provided node_project sample was also used for the logic of the controller, router, and many querying functions.

AI Declaration

AI was used for exporting our DDL.sql into Google Sheets, as well as debugging the project after the main project was done.

https://gemini.google.com/share/fea2c4f4aeae

ASSETS								
TickerSymbol	Name	LastPrice	CurrentPrice	Exchange	LastUpdate	IsActive	Currency	CreatedDate
AAPL	Apple Inc.	180.5	182.1	NASDAQ	2026-03-10 10:0	1	USD	2020-01-15
MSFT	Microsoft Corp.	410.2	412.35	NASDAQ	2026-03-10 10:0	1	USD	2019-06-20
TSLA	Tesla Inc.	195.1	190.75	NASDAQ	2026-03-10 10:0	1	USD	2021-03-08
SHOP	Shopify Inc.	78.4	79.25	NYSE	2026-03-10 10:0	1	CAD	2020-09-01
NVDA	NVIDIA Corp.	870	865.5	NASDAQ	2026-03-10 10:0	1	USD	2018-11-12
BB	BlackBerry Ltd.	5	4.8	NYSE	2026-03-10 10:0	0	USD	2010-01-01
JPM	JPMorgan Chase	190	192.5	NYSE	2026-03-10 10:0	1	USD	2005-05-15
STOCKS								
TickerSymbol	Industry	Sector	SharesOutstandi	EarningsPerShare				
AAPL	Electronics	Technology	1,000,000,000	6.25				
MSFT	Software	Technology	2,000,000,000	11.5				
TSLA	Automotive	Consumer	3,000,000,000	3.1				
SHOP	E-Commerce	Technology	4,000,000,000	0.9				
NVDA	Semiconductor	Technology	5,000,000,000	14.75				
BB	Software	Technology	500,000,000	-0.5				
JPM	Banking	Finance	2,800,000,000	16.2				
PORTFOLIO								
CreatedDate	Name	Currency						
2026-01-01	Growth	USD						
2026-01-15	Value Picks	USD						
2026-02-01	Canadian	CAD						
2026-02-15	Volatile	USD						
2026-03-01	Dividends	USD						
PEOPLE								
Email	PhoneNumber	Networth	Name	Cash				
a@demo.com	777-777-7777	250,000.00	Alice Test	15,000.00				
b@demo.com	777-777-7778	120,000.00	Bob Tester	8,000.00				

c@demo.com	777-777-7779	500,000.00	Caleb Testerer	25,000.00				
d@demo.com	777-777-7780	66,666.00	Dylan Caleb	6,000.00				
e@demo.com	777-777-7781	300,000.00	Edward Bob	12,000.00				
PEOPLE HAS PORTFOLIO								
CreatedDate	Email							
2026-01-01	a@demo.com							
2026-01-15	b@demo.com							
2026-02-01	c@demo.com							
2026-02-15	d@demo.com							
2026-03-01	e@demo.com							
CONTAINS								
Email	CreatedDate	TickerSymbol	Quantity	PurchasePrice				
a@demo.com	2026-01-01	AAPL	20	175				
a@demo.com	2026-01-01	TSLA	30	190				
b@demo.com	2026-01-15	MSFT	10	395				
c@demo.com	2026-02-01	TSLA	15	205				
d@demo.com	2026-02-15	SHOP	30	75				
e@demo.com	2026-03-01	NVDA	5	820				
COMPANIES								
CompanyID	FoundedDate	IsPublic	EmployeeCount					
101	1976-04-01	1	160,000					
102	1975-04-04	1	220,000					
103	2003-07-01	1	140,000					
104	2006-09-01	1	12,000					
105	1993-04-05	1	30,000					
106	1993-05-05	0	30,000					
107	2000-12-31	1	250,000					

OWNS								
Email	CompanyID	Percentage						
a@demo.com	101	20						
a@demo.com	102	20						
a@demo.com	103	20						
a@demo.com	104	20						
a@demo.com	105	20						
a@demo.com	106	50						
a@demo.com	107	10						
b@demo.com	102	0.75						
c@demo.com	101	1						
c@demo.com	102	1						
c@demo.com	103	2.25						
c@demo.com	104	1						
c@demo.com	105	1						
d@demo.com	104	0.5						
e@demo.com	105	1.1						
TRADERS								
Email	Performance	AverageTradeTime						
a@demo.com	8	12						
b@demo.com	5	25						
c@demo.com	9	8						
d@demo.com	4	30						
e@demo.com	7	15						
CEOS								
Email	InCompanyTime	PublicConsensus	PreviousRoles					
a@demo.com	6	72	CFO					
b@demo.com	3	55	Founder					
c@demo.com	9	81	CEO					

d@demo.com	2	40	Finance Director					
e@demo.com	5	68	Product Manager					
REPORTER								
Email	PublicConsensus							
a@demo.com	70							
b@demo.com	52							
c@demo.com	85							
d@demo.com	45							
e@demo.com	66							
TIMEFRAME								
TimeFrameID	Volume	Time_to_time	AverageChange					
1	5,000,000	1D	2					
2	8,000,000	1W	-1					
3	12,000,000	1M	5					
4	3,000,000	3M	7					
5	15,000,000	1Y	12					
DAYS								
TimeFrameID	DayOfMonth	DayOfWeek	IsHoliday	Has_Months				
1	10	Mon	0	3				
2	11	Tue	0	3				
3	12	Wed	0	3				
4	13	Thu	0	3				
5	14	Fri	0	3				
MONTHS								
TimeFrameID	MonthOfYear	Has_Years						
1	3	2026						
2	3	2026						

3	2	2026						
4	1	2026						
5	12	2025						
YEARS								
TimeFrameID	YearNumber							
1	2026							
2	2026							
3	2026							
4	2026							
5	2025							
NEWS EVENTS								
EventID	Headline	SentimentScore	TimeFrameID	Happened				
201	Earnings Beat	85	1	1				
202	Product Recall R	30	2	1				
203	Regulatory Appr	78	3	1				
204	CEO Interview	60	4	1				
205	Market Selloff	20	5	1				
BUYS								
Email	TickerSymbol	Quantity						
a@demo.com	AAPL	5						
b@demo.com	MSFT	2						
c@demo.com	TSLA	3						
d@demo.com	SHOP	10						
e@demo.com	NVDA	1						
b@demo.com	TSLA	15						
b@demo.com	JPM	50						
d@demo.com	AAPL	100						
d@demo.com	NVDA	20						

SELLS								
Email	TickerSymbol	Quantity						
a@demo.com	AAPL	1						
b@demo.com	MSFT	1						
c@demo.com	TSLA	1						
d@demo.com	SHOP	2						
e@demo.com	NVDA	1						
TRADES								
TickerSymbol	CompanyID	PublicDate	PercentageOfCompany					
AAPL	101	2026-01-20	10					
MSFT	102	2026-01-22	12					
TSLA	103	2026-02-05	15					
SHOP	104	2026-02-18	8					
NVDA	105	2026-03-05	20					
OPERATES								
Email	CompanyID	IsFounder	InOfficeTime					
a@demo.com	101	0	6					
b@demo.com	102	0	3					
c@demo.com	103	1	9					
d@demo.com	104	0	2					
e@demo.com	105	0	5					
IMPACTS								
TickerSymbol	EventID	PreEventPrice	PostEventPrice					
AAPL	201	180	185					
MSFT	202	410	405					
TSLA	203	195	205					
SHOP	204	78	80					

NVDA	205	870	845					
ON RELATION								
TickerSymbol	EventID							
AAPL	201							
MSFT	202							
TSLA	203							
SHOP	204							
NVDA	205							
REPORTS								
Email	EventID							
a@demo.com	201							
b@demo.com	202							
c@demo.com	203							
d@demo.com	204							
e@demo.com	205							